School of Engineering | Lancaster University

INDIVIDUAL MEng PROJECT VIVA Presentation

‘Investigating the feasibility of energy extraction from the deflection of aerofoils using the aerodynamic phenomenon of vortex shedding’

James Roberts

Supervisor: Stephen Quayle

May 2026

Project Motivation and Problem Context

- **UK net-zero targets** - (100% Zero Carbon Generation by 2030 and net-zero by 2050).
- **Limitations of traditional wind turbines** - (threat to bird population and hysteresis in high wind load conditions ~ 25m/s).
- **Lack of research into generating energy from VIV** (Vortex Induced Vibration) devices, especially on a small scale.
- There is an opportunity and interest in this field as the EU Commission has invested €1,328,687.50 into similar projects conducted by Vortex Bladeless.

School of Engineering | Lancaster University

SOLIDWORKS

Aims and Objectives

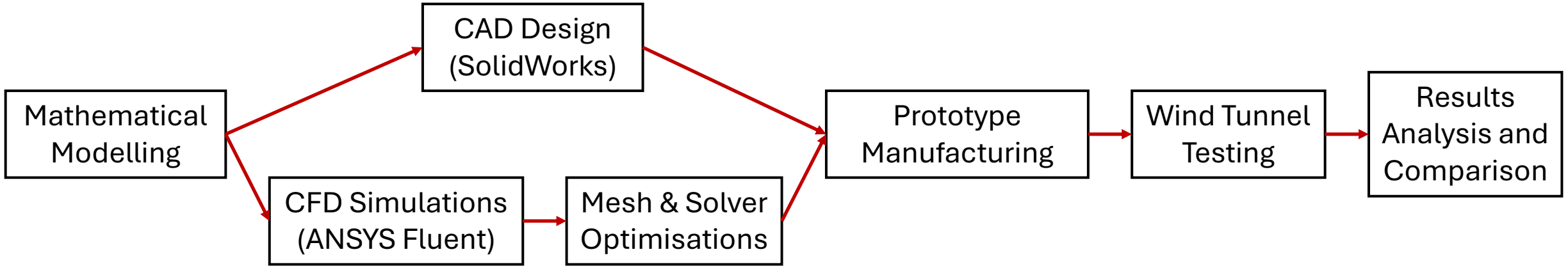
Ansys FLUENT

- **Prove the feasibility of the VIV device concept as a method of energy generation.**
- Use CAD & CFD tools to develop a small-scale model of a VIV device fit for functionality within the wind tunnel.
- Create methods for data extraction when testing the physical model of the VIV device.
- Analyse and compare the predicted behaviour to the observed behaviour of the VIV device.
- Identify the applications for which such device is suitable for.

School of Engineering | Lancaster University

Methodology

- The workflow of the project followed the general progression as shown below:

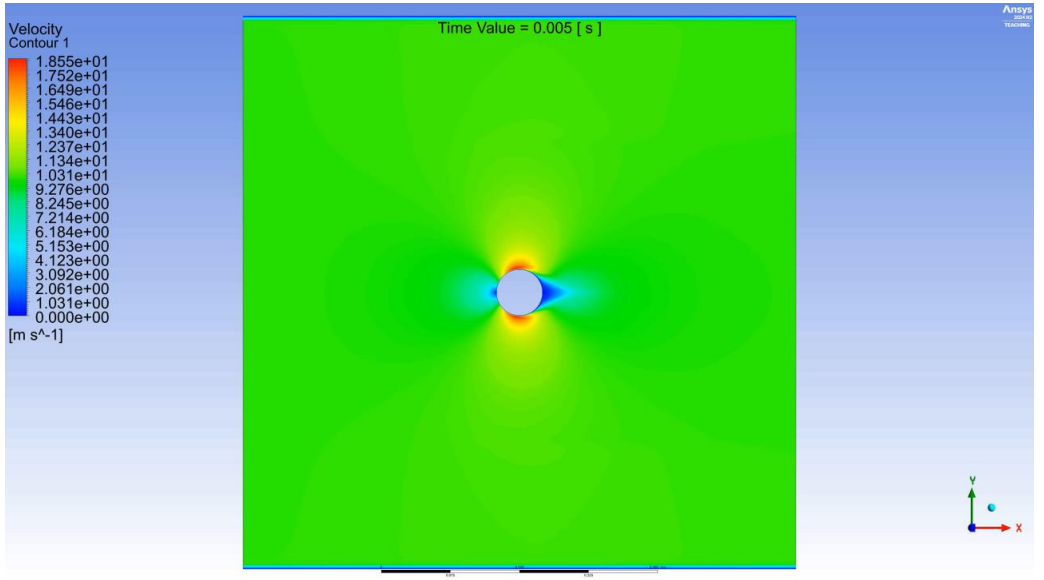


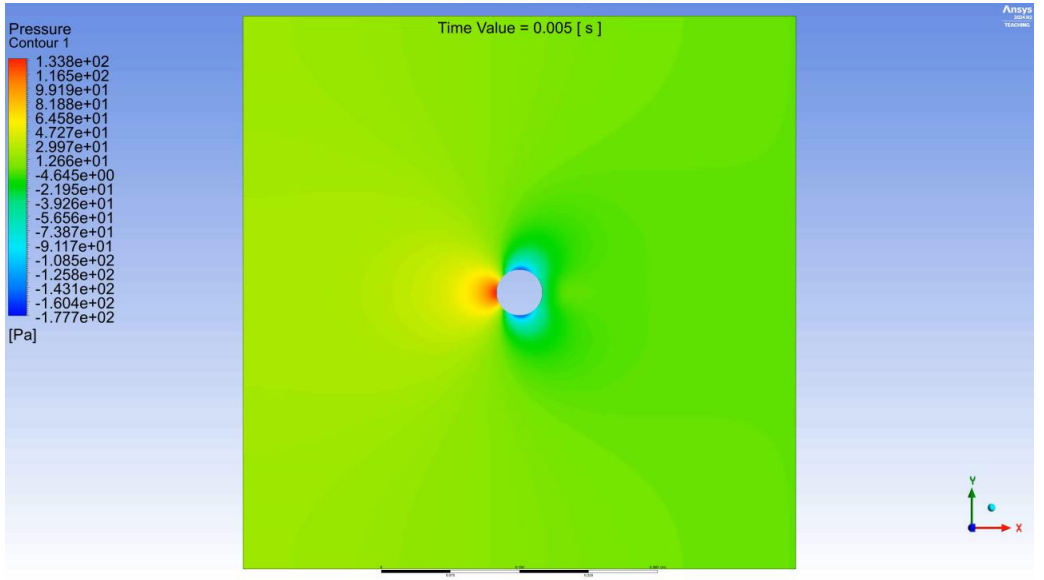
School of Engineering | Lancaster University

Ansys FLUENT

Scientific Principles and Theorems

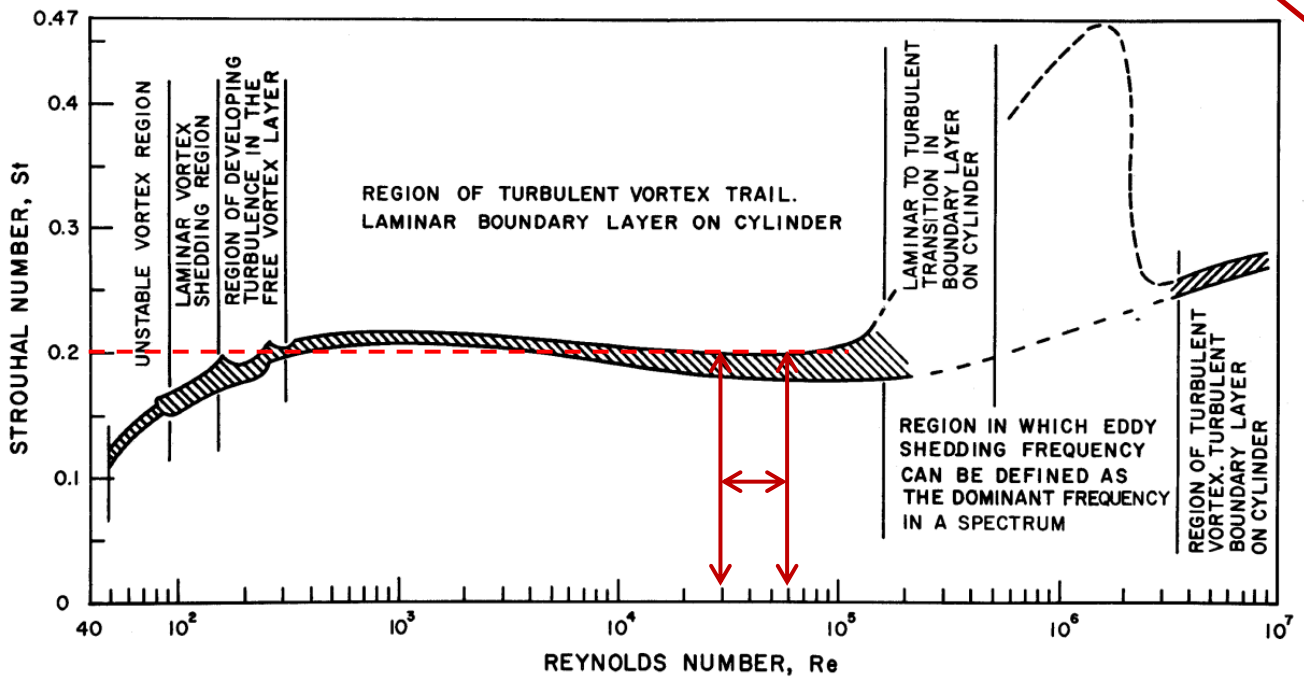
- **Reynolds number** of a cylinder is: $Re = \frac{U \cdot D}{\nu}$, Determines the flow regime
- **Strouhal equation** for vortex shedding frequency is: $f = \frac{St \cdot U}{D}$, Strouhal number predicts shedding frequency ~ 0.2.
- **Vortex Induced Lock In** – (Where shedding frequency = natural frequency of the structure).
- **Structural analysis** on the metal rod structure using classic beam theory.





Scientific Principles and Theorems

Maximum and minimum Re values tested with 50mm diameter at 20m/s and 10m/s where: 6.76×10^4 *and* 3.38×10^4.



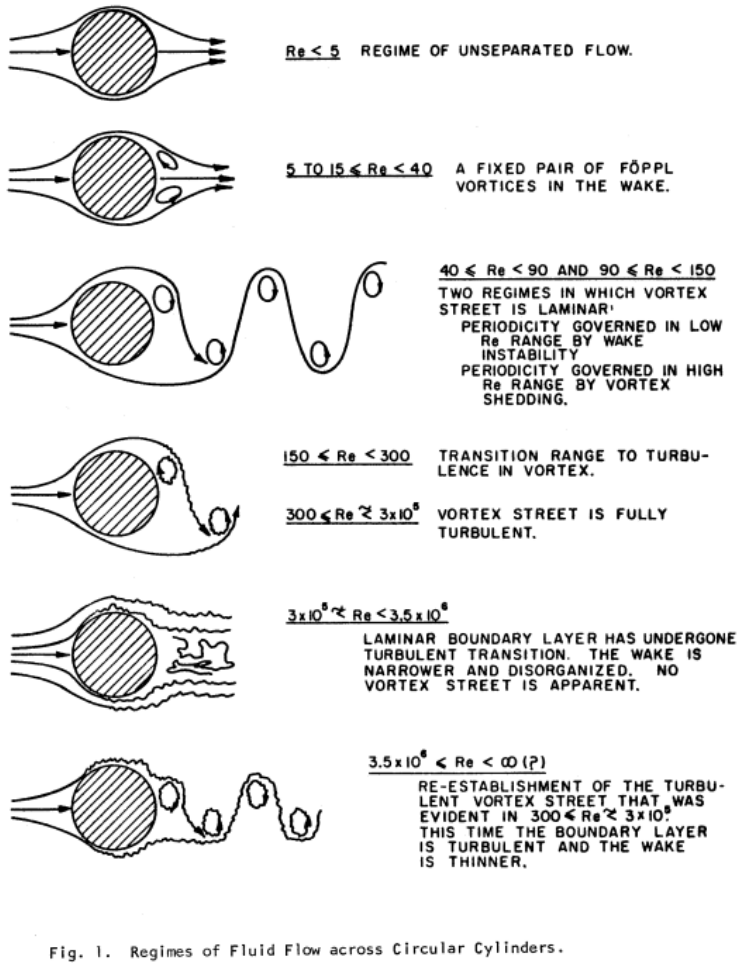


Fig. 1. Regimes of Fluid Flow across Circular Cylinders.

School of Engineering | Lancaster University

CFD Setup and Key Decisions

Ansys FLUENT

- **Square boundaries** to represent the wind tunnels boundary conditions with inlet and outlet on opposing boundaries.
- **Infiltration layers** added to surround the cylinder to resolve the transient boundary layer with a greater resolution. Which yielded an **average y^+ value of 1.13 - 2.d.p**.
- **Time step of 0.0005s** to accurately capture a predicted max 80Hz shedding frequency within the solver.
- **k-ω SST turbulence model** used as it best resolves turbulent boundary layers.
- **Cl, Force and Shedding Frequency** were variables extracted from Fluent for analysis.

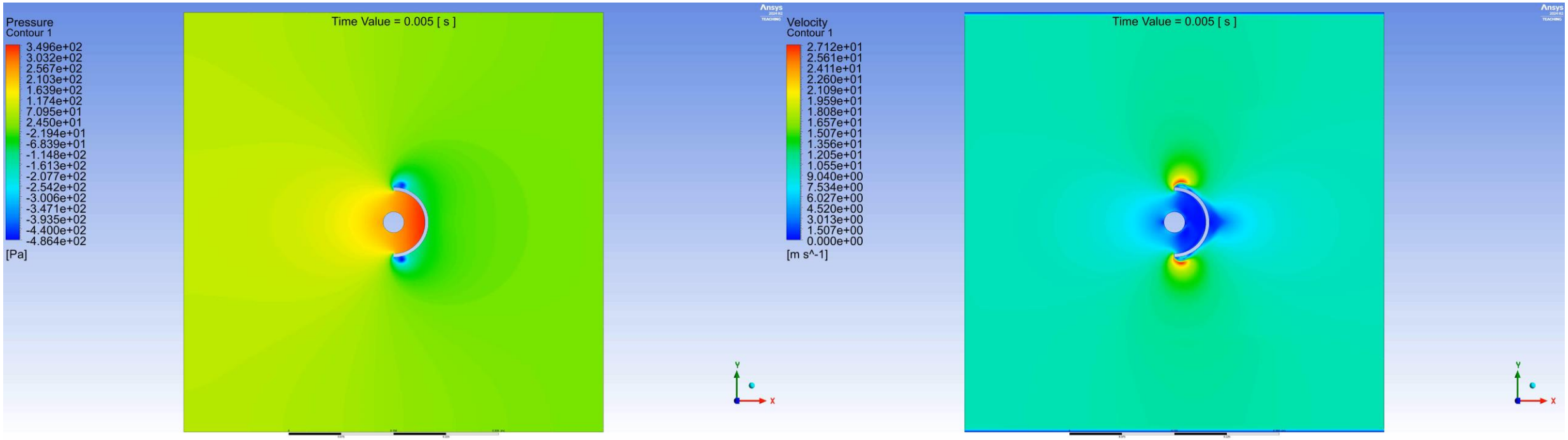
School of Engineering | Lancaster University

Ansys FLUENT

Amplifier Design

The amplifier device operates by generating larger vortices at greater angles compared to that of conventional vortex shedding, to the detriment of a greatly reduced shedding frequency

10m/s fluid flow with amp cross-section dimensions



Manufacturing and Practical Constraints

- The height of the VIV device was limited to the height of the **wind tunnel boundaries at 600mm**.
- **Wind tunnel speed variations**, especially at lower velocities.
- The model P3 Strain Indicator and Recorder had a **sample rate of 1Hz**, resulting in a great amount of lost data points during oscillations and peak points during oscillations.





School of Engineering | Lancaster University

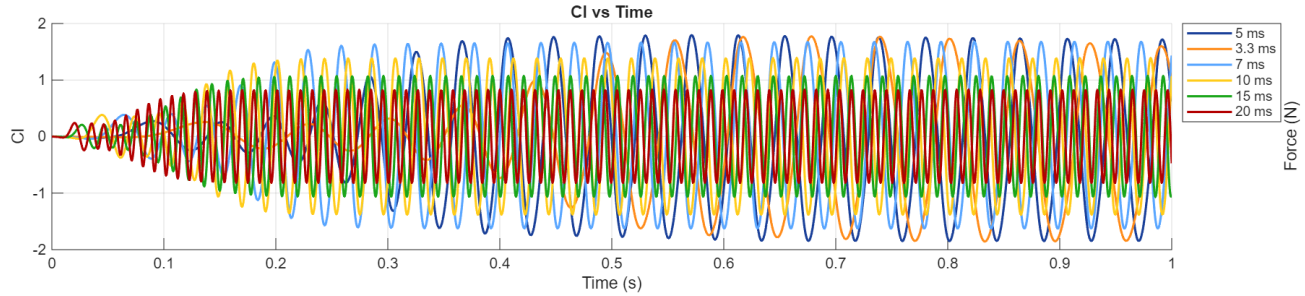
Wind Tunnel Testing

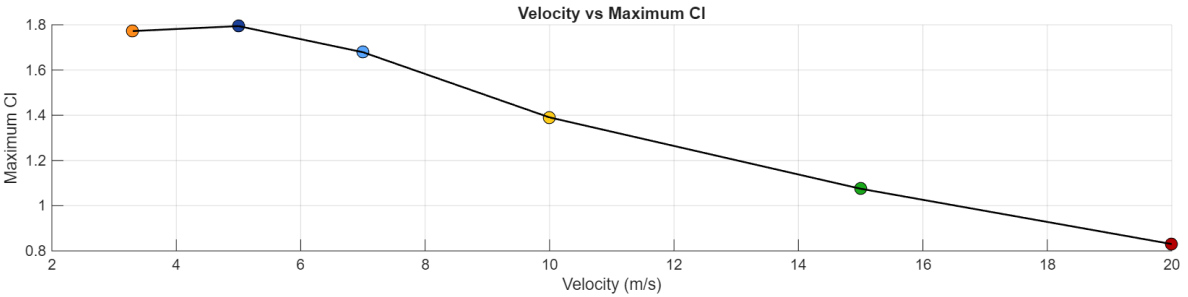
3 different recording methods trailed:

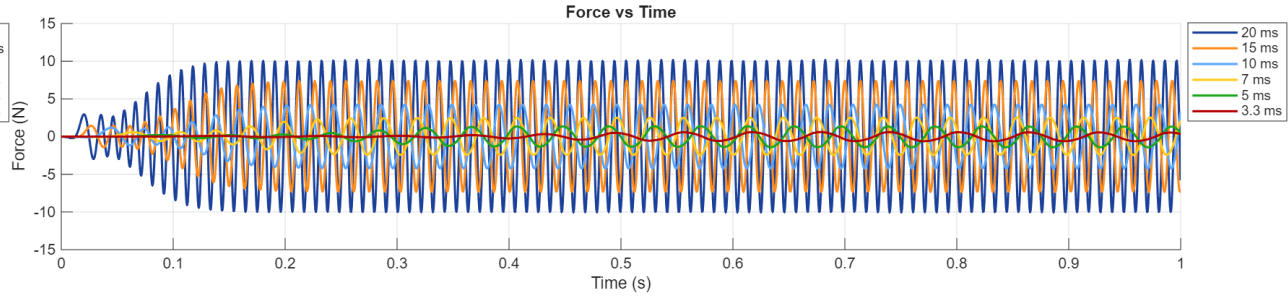
- Camera Mounting and Visual Aids.
- Digital Recordings from the P3 Strain Indicator and Recorder using the half bridge strain gauge configuration.
- Using the P3 in analogue mode with a coaxial cable in conjunction with an oscilloscope.

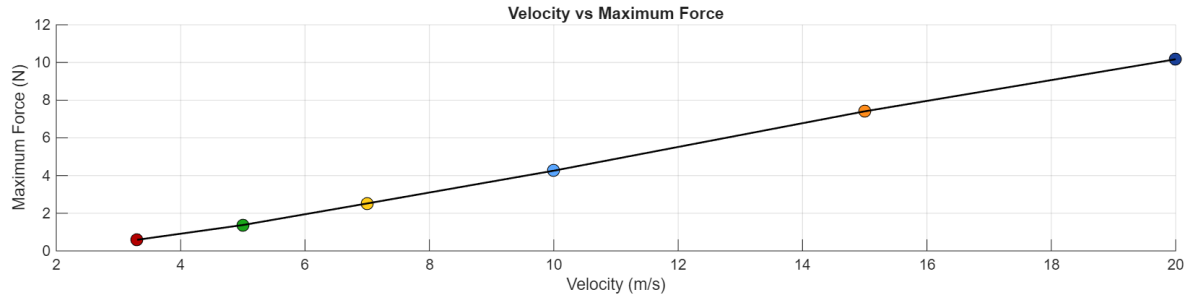


Results Comparison (Predicted)









CFD Results when using 50mm Diameter.

Note – Force is per meter length of the aerofoil

School of Engineering | Lancaster University

MICRO MEASUREMENTS A VPG Brand

Results Comparison (Actual)

Analgoue Oscilloscope Data				Digital P3 Data
Tip Type	Velocity	Peak Force (N)	Oscillation Frequency (Hz)	Average Force (N)
DOME	10m/s	0.53940	7.5	0.04613
	15m/s	0.71920	7.5	0.07803
	20m/s	1.43840	7.0	0.15900
TAPER	10m/s	0.53940	7.5	0.04886
	15m/s	0.89900	7.5	0.05247
	20m/s	1.07520	7.5	0.14510
EXPANDING	10m/s	0.43152	7.0	0.04109
	15m/s	0.46748	7.0	0.04600
	20m/s	0.89900	7.0	0.08120
CONTROL	10m/s	**0.32364**	6.0	0.04763
	15m/s	**0.35960**	7.0	0.11494
	20m/s	**1.22264**	6.0	0.17402
AMPLIFIED	3-4m/s	0.35960	5.0	0.07407
	7m/s	0.82708	6.0	0.13930
	10m/s	1.15072	5.0	0.19100
	15m/s	2.15760	5.5	0.50888
	20m/s	7.19200	6.5	0.92628

From the predicted values (aerofoil length is 500mm) we'd expect peak forces of:

- 10m/s ≈ 2.1N
- 15m/s ≈ 3.75N
- 20m/s ≈ 5.0N

Instead, the direct comparison of the control test showcases that **less force was generated than predicted.**

Furthermore, the **amplifier device** increased the peak force of the oscillation to **7.2N greater than the CFD** suggests for a clean conventional vortex shedding surface.

Feasibility and Application

- **VIV generation is feasible but niche at this stage**.
- The current design is best suited for rural low energy sensor powering applications.
- Potential in the design to progress the effectiveness forward by improving the fundamental design.
- **Omnidirectionality**.

Scalability greatly improves the efficiency as:

The lift force generated from peak vortex shedding is directly proportional to the radius and length of the cylinder. As shown by: $F_L = \frac{1}{2}\rho v^2(2r \cdot L)C_l$. If we **scale r and L there's a 4x increase in lift**.

School of Engineering | Lancaster University

Engineering Management and Reflection

- Development of the **Gantt Chart** provided clear and attainable goals throughout the project ensuring that the project stayed on schedule.
- **Risk matrix** was created and the **wind tunnel risk assessment** was followed ensuring safe and reliable operation during experimentation and the project.
- Each mistake and lesson will be taken forward, **improving my project management and problem-solving skills** making me adapted for real life workflows in my career.

Risk No.	Risk Description	Likelihood 1 to 5	Impact 1 to 5	Severity 1 to 25	Mitigation	Contigency
1	Supply Constraints	3	3	16	Ensure that any parts that ive ordered / developed are done well in advance of dealines.	Use alternate materials and parts from other distributers. Furthermore use in house items that the uni already has.
2	Lack of Wind Tunnel Availability	4	3	14	Intend to make my scale model as early as physically posable to avoid conflicts with other students using the one wind tunnel. This will allow for time sceduling for who is using the wind tunnel and when.	Resort to real life testing if the weather is windy, if not wait for a slot to fit my experiment in or use simular machines that can generate a steady flow of air.
3	Smal Technician Workforce	5	2	6	Ensure any drawings are submitted with ample time in hand and learn how to use required machines before hand to reduce workload on the staff.	Create my parts out of methods that im capable of such as 3D printing and learn how to use the wind tunnel beforehand.
4	Student Illness	3	2	7	Maintain healthy practices to minimize the chances of falling ill. Don't take any dangerous actions that could cause harm to oneself.	Work remotley using MyLab on the digital aspects of the project such as CAD and CFD work. Test my scale model if possible in an isolated group if possible to prevent the spread of disease.
5	Data Loss Due to Software Crashes	1	4	21	Use cloud based storage and auto save features so that if any fatal crashes occur the data is backed up preventing data loss. Save files constantly if auto save isnt available and upload to OneDrive.	Redo the work best from memory, see if there is data recovery. Use a log book to note down each step you taked and use that to redo the task at hand.
6	Simulation work is limited due to academic license	5	2	6	If possible only simulate one side of the aerofoil as it should be symertric to reduce the amount of mesh divisions used.	Make do with a less acurate simulation from CFD and take it as an idication as to its behaviour, use as many mesh divisions as possible.
7	Significant events out of my control (Pandemic / Building closures)	1	5	23	Stay on the project timeline as much as possible to avoid distruption, get everything built and developed as early as possible. Take advice from professionals on how to procide.	Alter the scope of the project if the affect is significant to the progression of the investigtion. Carry on as best a physically possible.
8	Break the Budget	1	1	12	Ensure everything is correct and the cost is tallied up properly before any orders are made, make sure the designs are final and have confidence in them to avoid rebuilds.	Don't build anymore use what ive got at that point, increase the theoretic aspects of the project, conduct more CFD runs. Try to fix what ive got and test it non the less to get results.
9	Superviser Unavailable	2	1	6	Keep regular contact with my superviser even if that's online via a teams / zoom meeting.	Complete as much work possible without the need of the superviser, use communications such as email and other online platforms to keep in contact with the superviser if they're required.
10	Fall Behind Schedule	3	4	17	Allow for tasks to take longer than expected with buffer periods and time for redevelopment.	Make more time available from my project from weekends and make it my sole priority with other uni work, find more efficient ways of working or work on another aspect on the project if a certain bit is taking too long.

Conclusions

- **The project has proved VIV devices as a feasible method of energy generation.**
- CDF validated trends expected from the mathematical modelling.
- **Amplifer increased the magnitude of oscillations significantly.**
- **Forces without the amplifier small than expected.**
- Further research into this field would be necessary to generate a design suitable for applicable energy generation.
- Less energy dense than conventional wind turbines.

Recommendations for Future Work

- Devices in **series and parallel**, examine how they affect each other through the generated turbulence.
- Further **refine the amplifier design** by changing variables such as: Basket diameter, height and location of the amplifier.
- **Research into using dead aerodynamic structures** such as electrical lines and lampposts to generate electricity can be another substitution rather than dedicated generators.
- Further **scalability** studies
- **Underwater analysis** for the device, determine whether its better suited for air or water.
- **Methods for energy extraction**, Eg: Magnetic solution using Faraday's Law and electromagnetic induction or a mechanical solution using a cam and gearbox system connected to a turbine.

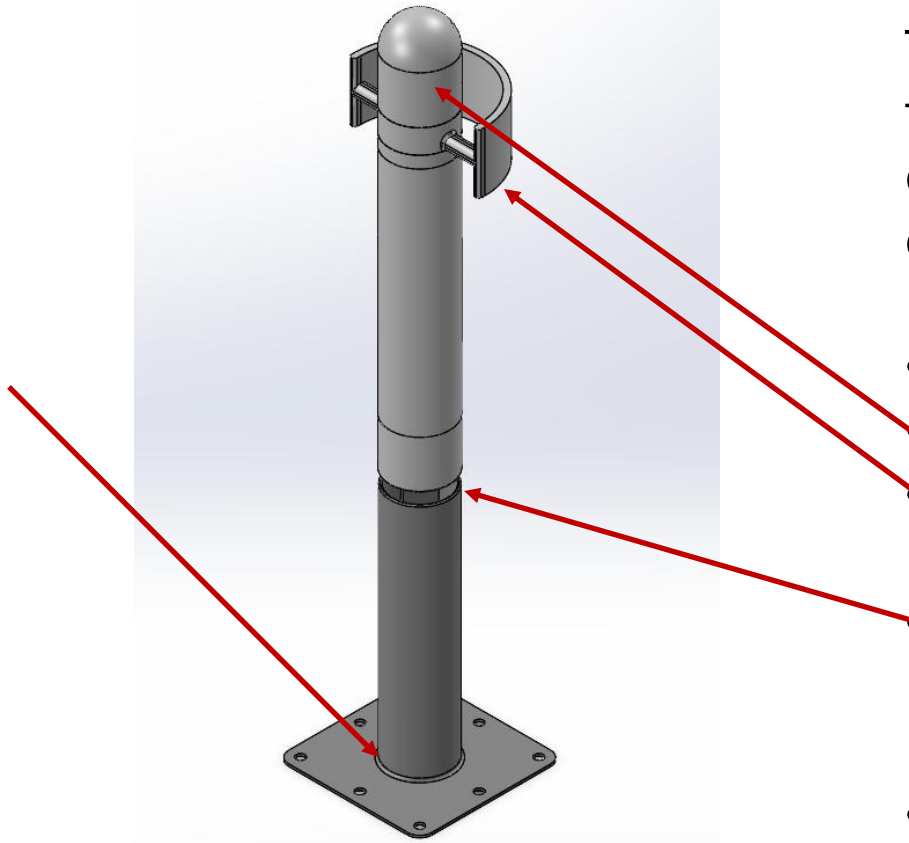
Q&A

School of Engineering | Lancaster University

SOLIDWORKS

Incorporating Technical Insights Developed Throughout this Project

Bottom of aerofoil connected to the baseplate to produce a constant wake field



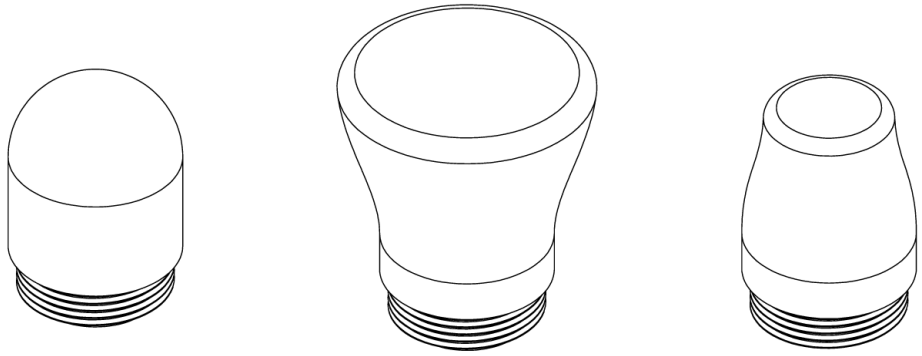
The design through experimentation that theoretically would provide the greatest efficiency in generating energy would consist of:

- Uniform 50mm Diameter.
- Dome tip at the top.
- Amplifer device scaled up to a 50mm diameter.
- Separation of the bottom of the aerofoil to the load bearing structure (due to cantilever effect).
- 6.5Hz designed shedding frequency for resonance.

Device Design and Engineering Decisions

- 50mm diameter chosen as it provided the optimum compromise when comparing force and shedding frequency.

Due to the tip containing the largest bending moment (M=Fd) and the greatest region of power generation due to (P=Fv).



- 3 Tip designs were developed to determine whether increasing force or shedding frequency increases the devices.

School of Engineering | Lancaster University

Ansys FLUENT

Conventional Vortex Shedding For Comparison

50mm Diameter at 10m/s fluid flow:

